

DC Motor Speed Control with a PID Controller

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Control Systems & Simulation · MATLAB

Overview

This project designs and simulates a closed-loop speed controller for a brushed DC motor in MATLAB. Open-loop drive — applying a fixed voltage — cannot hold a target speed once the load changes, so a **PID (proportional-integral-derivative)** controller is used to continuously correct the motor voltage from the speed error. The same control problem appears throughout semiconductor process and test equipment, where PID loops regulate furnace temperature, chamber pressure, and wafer-handling motion. The model is built from first principles and runs in base MATLAB with no toolboxes.

Plant Model

The motor is represented by its coupled electrical and mechanical equations: the armature circuit ($V = R \cdot i + L \cdot di/dt + K_e \cdot \omega$) and the rotor dynamics ($J \cdot d\omega/dt = K_t \cdot i - b \cdot \omega - T_{load}$). These are integrated numerically over time, giving a realistic plant that includes the electrical and mechanical time constants and responds to an external load torque.

Controller Design

- **Discrete PID** running at a 1 kHz sample rate: the speed error drives proportional, integral, and derivative terms that sum to the commanded voltage.
- **Output saturation** clamps the command to the ± 12 V supply limit, and **integral anti-windup** stops the integrator from accumulating while saturated — preventing large overshoot after aggressive transients.
- **Gain tuning** ($K_p=0.4$, $K_i=25$, $K_d=0.002$) balances a fast rise against low overshoot while guaranteeing zero steady-state error from the integral term.

Results

Against a 1000 rpm setpoint the closed-loop response achieves the metrics below. A step load torque applied at 0.5 s lets the controller be compared directly to open-loop drive: the PID loop dips only $\sim 1\%$ and returns to setpoint, while the open-loop motor droops toward ~ 860 rpm and never recovers (Figure 1).

Overshoot	Rise time (10–90%)	Settling (2%)	Steady-state error	Disturbance dip
6.2 %	38 ms	96 ms	0 rpm	1.0 % (recovers)

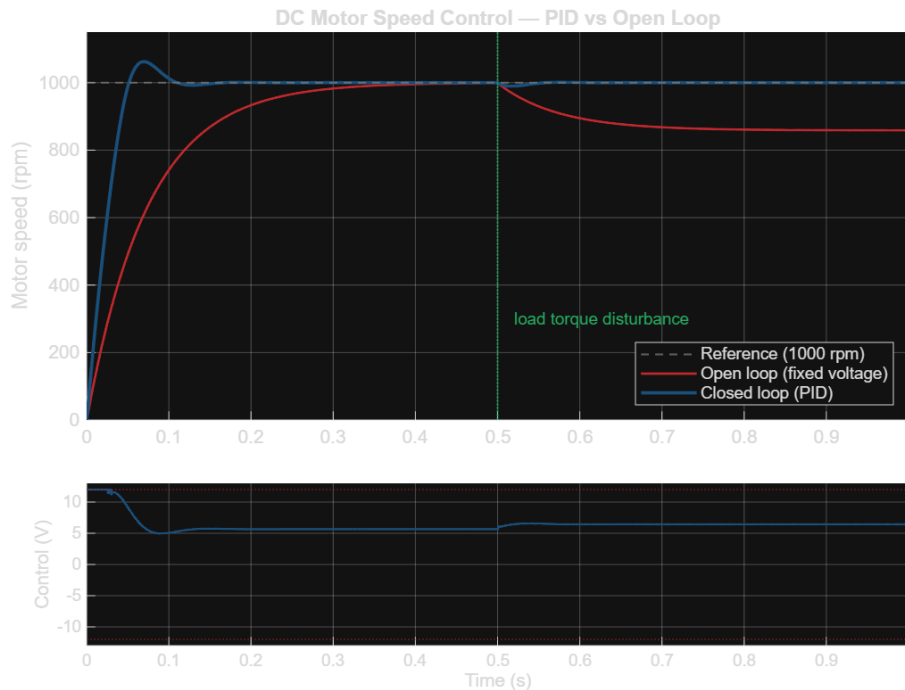


Figure 1. MATLAB simulation. Top: motor speed for open-loop (red) vs PID (blue) against the 1000 rpm reference, with a load-torque disturbance at 0.5 s. Bottom: PID control voltage, saturating at 12 V during the initial transient and rising to reject the disturbance.

Tools & Skills Demonstrated

Control-system modeling of coupled electromechanical dynamics; discrete-time PID design, tuning, saturation, and anti-windup; step-response and disturbance-rejection analysis; numerical simulation and plotting in MATLAB.